

Homework 2

ASEN 5335 Aerospace Environment

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Problem 1: Spaceballs

(i.)

We are able to use the provided MSISatmosphere1000.m profile to calculate the an approximate total mass of the atmosphere. The MSISatmosphere1000 function takes inputs of altitudes and outputs a structure, one of which is mass density which corresponds to that altitude. Using MATLAB we are able to have a vector of altitudes that range to 1000 km, the maximum model input of the MSISatmosphere1000 function. We can then obtain a vector of mass densities ρ . We use the following,

$$m = \rho V$$

$$m = \int \rho dV$$

And for a sphere,

$$m = \int \rho * 4\pi r^2 dr$$

$$\therefore m = 4\pi \int \rho r^2 dr$$

With ρ being a vector and r also being a vector of radius of this atmospheric shell (altitude including radius of the Earth $R_E = 6371$ km), we can integrate within MATLAB to obtain a

$$\boxed{m = 5.297 * 10^{18} \text{ kg}}.$$

(ii.)

Assuming a volume of $V = 10 * 10^3 * 10 * 10^3 * 100 * 10^3 = 1 * 10^{13} \text{ m}^3$, a constant temperature $T = 300 \text{ K}$, molar mass $M = 0.029 \text{ kg/mol}$, and universal gas constant $R = 8.314 \text{ J/(molK)}$. We can use the ideal gas law,

$$PV = \frac{m}{M}$$

Rewrite to solve for pressure P ,

$$P = \frac{m}{MV} RT$$

$$\boxed{P = 4.556 * 10^{10} \text{ Pa}}$$

Problem 2: Atmospheric Scale Height

To calculate the scale height H of air mixture assuming N_2/O_2 mixture at sea level, we use the equation.

$$H = \frac{kT}{mg}$$

Where we assume a constant $T = 300$ K from the problem, and $k = 1.38 * 10^{-23}$ J/K, the Boltzmann constant, $g = 9.81$ m/s², the gravitational constant. We then seek the molecular mass of N_2/O_2 mixture. This can be obtained from the table provided in the HW. We find the atomic mass of the air mixture provided with the % by volume

$$amu_{air} = 28.0134 (.781) + 31.9988 (.209) = 28.566 \text{ g/mol}$$

$$m_{air} = amu_{air} * \frac{1 \text{ kg}}{1000 \text{ g} * 6.02 * 10^{23} \text{ g/mol}}$$

$$m_{air} = 4.745 * 10^{-26} \text{ kg}$$

This process of finding molecular weight and converting it can be repeated for atomic Oxygen, helium, and hydrogen.

$$amu_{oxygen} = (31.9988 \text{ g/mol})/2 = 15.9994 \text{ g/mol}$$

$$amu_{helium} = 4.0026 \text{ g/mol}$$

$$amu_{hydrogen} = 2.01594 \text{ g/mol}$$

$$m_{oxygen} = 2.6577 * 10^{-26} \text{ kg}$$

$$m_{helium} = 6.6488 * 10^{-27} \text{ kg}$$

$$m_{hydrogen} = 3.3487 * 10^{-27} \text{ kg}$$

The scale height equation can be used with each respective molecular mass of the species in question.

$$H_{air} = 8.8938 * 10^3 \text{ m}$$

$$H_{oxygen} = 1.588 * 10^4 \text{ m}$$

$$H_{helium} = 6.347 * 10^4 \text{ m}$$

$$H_{hydrogen} = 1.2602 * 10^5 \text{ m}$$

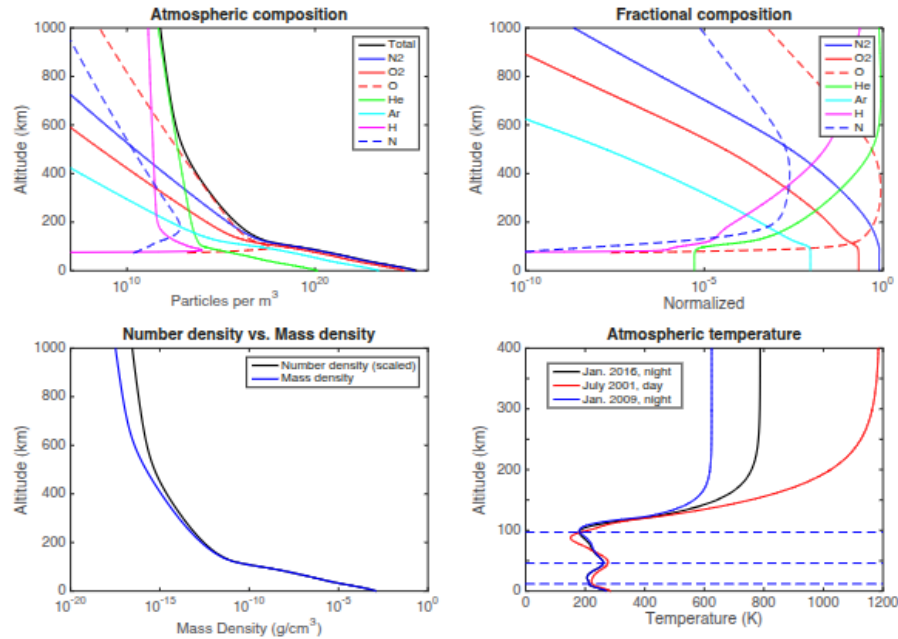


Figure 1: Atmospheric Profile

By comparing these scale heights to those in the Figure 1 provided from the HW, we observe that
 TODO: We seek to also calculate scale height from the MSIS profiles. From the MSIS profiles there will be different temperatures instead of the fixed T used earlier. Using altitudes of 150,180,450,500,600,1000km.

Problem 3: Atmosphere Variation

(i.)

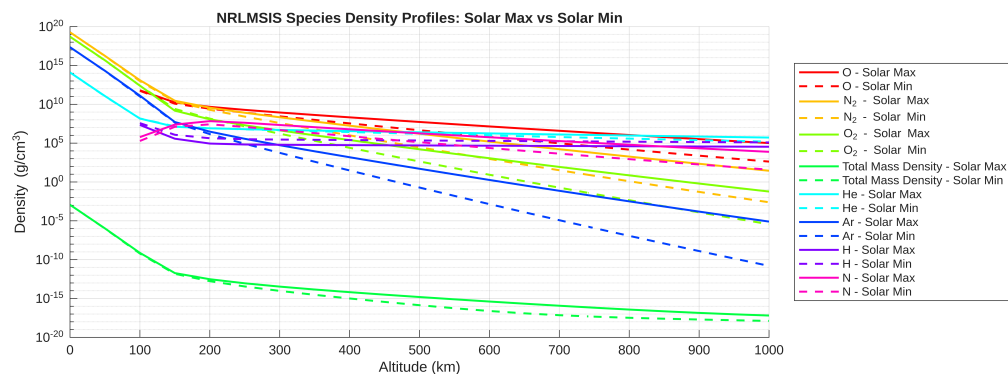


Figure 2: NRLMSIS Species Density Profiles Solar Max vs Solar Min

(ii.)

At 600 km, there is variation between the solar maximum and solar minimum with the density profiles of each species deviating. At solar maximums, the densities of each species tends to be larger by multiple orders of magnitude than they would be at solar minimums. This can even be seen overall with the Total Mass Density variation between solar maximums and minimums. The implication of this for spacecraft operations is that there will be different atmospheric conditions for spacecraft depending on if it is at a solar maximum or minimum. For instance, drag will be dependent on the density of the atmosphere, during solar maximums where density can increase by multiple orders of magnitude, drag will also be considerably affected. This could especially impact LEO orbits with their orbits being considerably influenced by drag. Their satellite operations could therefore vary depending on if it is at a solar maximum or minimum.

(iii.)

The dominant atmospheric constituent would be O, atomic oxygen, for both solar maximum and solar minimum. (TODO: Double check this...)

Problem 4: Oxygen Impact

(i.)

To find the characteristic velocity of an oxygen atom O at $T = 1000\text{ K}$ in the thermosphere, we are able to use the Maxwell Boltzmann Characteristic RMS Velocity equation.

$$v_{rms} = \sqrt{\frac{3k_B T}{m}}$$

$k_B = 1.38 * 10^{-23} \text{ J/K}$ as the Boltzmann constant, and we are able to find $m = 2.657 * 10^{-26} \text{ kg}$ for an oxygen atom. We can plug this into the equation to simply find,

$$v = 1248.26 \text{ m/s}$$

(ii.)

We then calculate the velocity that a spacecraft in LEO at 600 km would have,

$$\begin{aligned} v_{orbit} &= \sqrt{\frac{GM_{Earth}}{r}} \\ G &= 6.67 * 10^{-11} \text{ Nm}^2/\text{kg}^2 \\ M_{Earth} &= 5.972 * 10^{24} \text{ kg} \\ r &= 600 * 10^3 \text{ m} + 6371 * 10^3 \text{ m} = 6971 * 10^3 \text{ m} \\ \therefore v_{orbit} &= 7559 \text{ m/s} \end{aligned}$$

The energy imparted onto the spacecraft by an oxygen atom would be using the mass of the oxygen atom and the velocity of the spacecraft,

$$KE = \frac{1}{2} m_{\text{oxygen}} v_{\text{orbit}}^2$$

$$KE = 7.591 * 10^{-19} J$$

$$\text{Where } 1 J = 6.242 * 10^{18} \text{ eV}$$

$$\therefore KE = 4.738 \text{ eV}$$

(iii.)

TODO: TODO

Problem 5: Orbit Decay

(i.)

We use the following relations:

$$F_{\text{drag}} = \frac{1}{2} \rho V^2 C_D A, \quad V = \sqrt{\frac{GM_E}{a}}, \quad P^2 GM_E = 4\pi^2 a^3$$

Step 1: Drag equation

$$m \frac{dV}{dt} = \frac{1}{2} \rho V^2 C_D A$$

$$\frac{dV}{dt} = \frac{\rho V^2 C_D A}{2m}$$

Step 2: Relate da/dt to dV/dt with the velocity equation

$$V^2 a = GM_E$$

$$\left(\frac{d}{dt} \right) (V^2 a) = \left(\frac{d}{dt} \right) (GM_E)$$

$$2Va \frac{dV}{dt} + V^2 \frac{da}{dt} = 0$$

$$\frac{da}{dt} = -\frac{2a}{V} \frac{dV}{dt}$$

Substituting Step 1,

$$\frac{da}{dt} = -\frac{2a}{V} \left(\frac{\rho V^2 C_D A}{2m} \right)$$

$$\frac{da}{dt} = -\frac{a \rho V C_D A}{m}$$

Step 3: Relate da/dt to dP/dt with Kepler's third law

$$\begin{aligned}
 P^2 GM_E &= 4\pi^2 a^3 \\
 \left(\frac{d}{dt}\right) (P^2 GM_E) &= \left(\frac{d}{dt}\right) (4\pi^2 a^3) \\
 2P GM_E \frac{dP}{dt} &= 12\pi^2 a^2 \frac{da}{dt} \\
 \frac{da}{dt} &= \frac{P GM_E}{6\pi^2 a^2} \frac{dP}{dt}
 \end{aligned}$$

Step 4: Setting the two expressions for da/dt equal,

$$\begin{aligned}
 -\frac{a\rho VC_D A}{m} &= \frac{P GM_E}{6\pi^2 a^2} \frac{dP}{dt} \\
 \frac{dP}{dt} &= -\frac{6\pi^2 a^3 \rho VC_D A}{m P GM_E}
 \end{aligned}$$

Using $GM_E = V^2 a$ to eliminate GM_E ,

$$\begin{aligned}
 \frac{dP}{dt} &= -\frac{6\pi^2 a^3 \rho VC_D A}{m P V^2 a} \\
 \frac{dP}{dt} &= -\frac{6\pi^2 a^2 \rho C_D A}{m P V}
 \end{aligned}$$

For a circular orbit, $2\pi a = PV$,

$$\begin{aligned}
 \frac{dP}{dt} &= -\frac{6\pi^2 a^2}{m (2\pi a)} \rho C_D A \\
 \frac{dP}{dt} &= -\frac{3\pi a \rho C_D A}{m} \\
 \boxed{\frac{dP}{dt} = -3\pi a \rho \frac{C_D A}{m}}
 \end{aligned}$$

(ii.)

See section at end of assignment for full MATLAB Code.

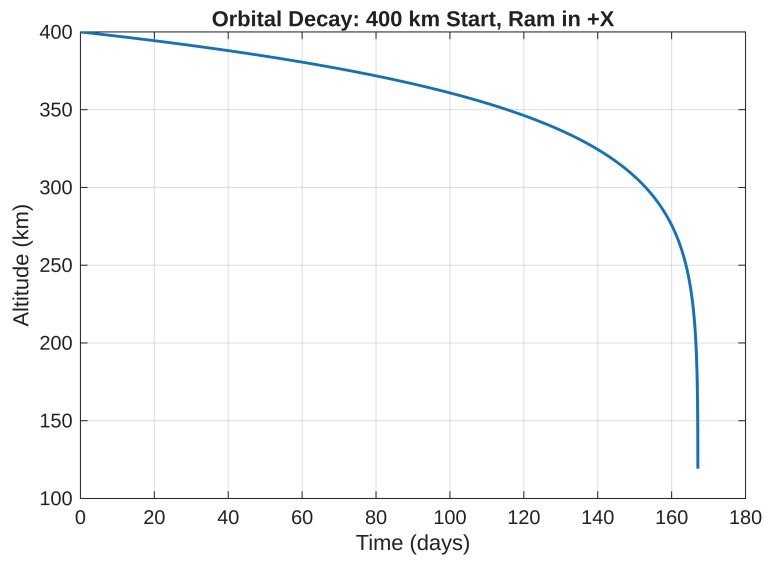


Figure 3: Orbital Decay for 400 km in +x

(iii.)

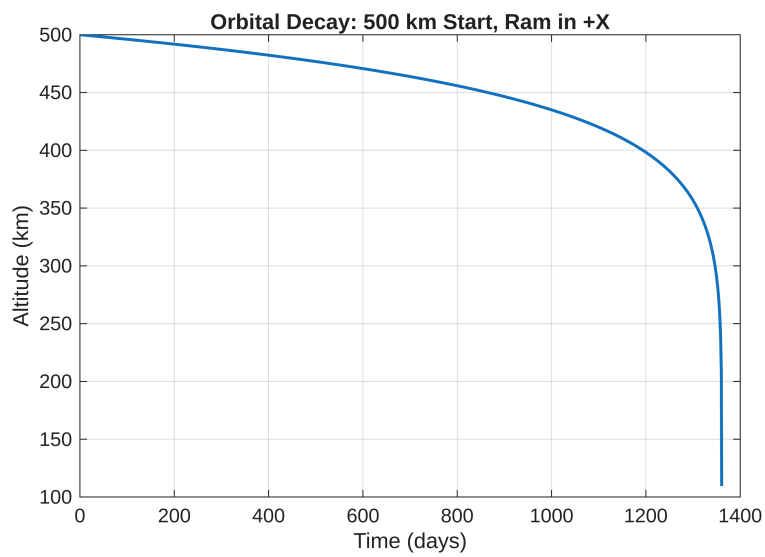


Figure 4: Orbital Decay for 500 km in +x

(iv.)

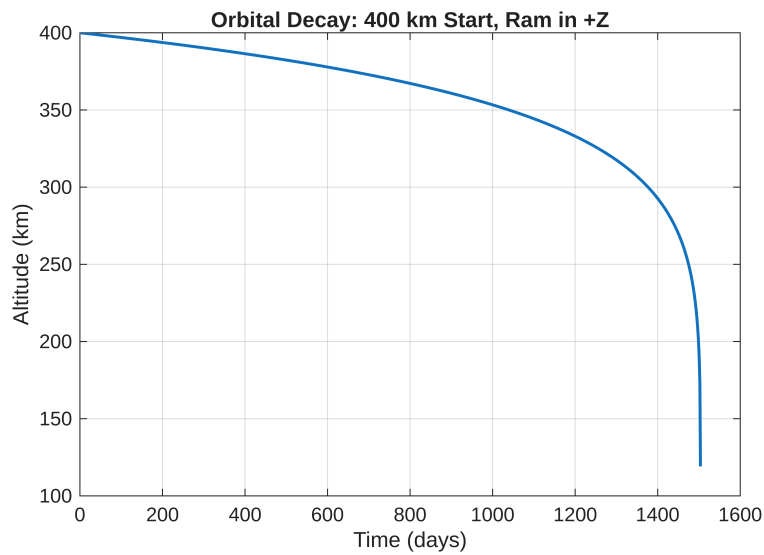


Figure 5: Orbital Decay for 400 km in +z

Problem 6: Atmosphere Uncertainty and Neutral Winds

(i.)

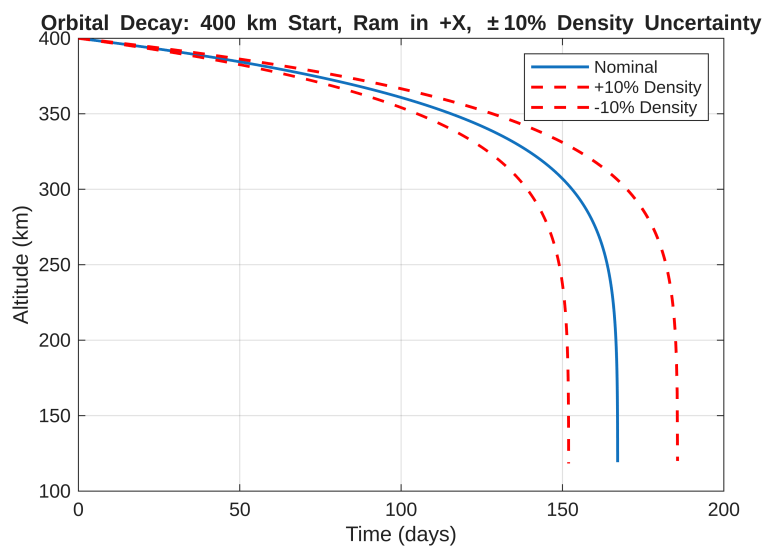


Figure 6: Orbital Decay for 400 km in +z with 10% Uncertainty

We are able to observe a difference in deorbit time of between 33 and 34 days based on the upper and lower bounds of density uncertainty.

(ii.)

Given a 90 minute orbit around Earth at some LEO altitude where this cubesat is operating, the implications for where this spacecraft re entry point in the atmosphere is that there is massive uncertainty in where it will re enter the atmosphere over it's entire ballistic deorbiting process. This location would greatly vary due to this uncertainty.

Problem 7: Air Breathing Thruster

(i.)

In order to find the thrust required to balance the drag force for some spacecraft operating at 400 km, with $C_D = 2.2$ and $A = 100 \text{ cm}^2 = 0.01 \text{ m}^2$, we would start with the drag equation,

$$F_D = \frac{1}{2} \rho V^2 C_D A$$

This means we seek to also solve for velocity and ρ

Solve velocity with the circular orbit relation from Kepler's Law

$$v = \sqrt{\frac{GM_E}{r}}, \text{ where } r = (400 * 10^3 \text{ m} + 6371 * 10^3 \text{ m})$$

$$v = 7700 \text{ m/s}$$

We can obtain the density at 400 km using the MATLAB function MSISatmosphere1000,

$$\rho = 1.194 * 10^{-15} \text{ g/cm}^3 = 1.194 * 10^{-12} \text{ kg/m}^3$$

$$\therefore F_D = 7.787 * 10^{-7} \text{ N} = 0.7787 \mu\text{N}$$

$$\boxed{\therefore F_T = F_D = 0.7787 \mu\text{N}}$$

(ii)

Now to find the exhaust velocity, we can use the relation,

$$F_D = -V_{ex} \frac{dm}{dt}$$

since $\frac{dm}{dt}$ must be equal to the mass collection rate, it is $\frac{dm}{dt} = \rho V A$, at the inlet

$$\therefore \frac{dm}{dt} = 9.194 * 10^{-11} \text{ m/s}$$

$$V_{ex} = -F_D / \frac{dm}{dt}$$

$$\boxed{V_{ex} = -8570 \text{ m/s}}$$

Problem 8: Missions

MATLAB Code

```

1  %% ASEN 5335 Aerospace Environment
2  %% Justin Le
3  %% HW2 Problem 3
4
5  clc;clear; close all
6
7  SOLAR_MAX = readmatrix('nrlmsis_output_SOLAR_MAX.txt');
8  SOLAR_MIN = readmatrix('nrlmsis_output_SOLAR_MIN.txt');
9
10 alt_max_km = SOLAR_MAX(:,6); % Height km
11 alt_min_km = SOLAR_MIN(:,6);
12
13 % Reading in species mass densities
14 O_max = SOLAR_MAX(:,9);
15 N2_max = SOLAR_MAX(:,10);
16 O2_max = SOLAR_MAX(:,11);
17 total_mass_max = SOLAR_MAX(:,12);
18 He_max = SOLAR_MAX(:,15);
19 Ar_max = SOLAR_MAX(:,16);
20 H_max = SOLAR_MAX(:,17);
21 N_max = SOLAR_MAX(:,18);
22
23 O_min = SOLAR_MIN(:,9);
24 N2_min = SOLAR_MIN(:,10);
25 O2_min = SOLAR_MIN(:,11);
26 total_mass_min = SOLAR_MIN(:,12);
27 He_min = SOLAR_MIN(:,15);
28 Ar_min = SOLAR_MIN(:,16);
29 H_min = SOLAR_MIN(:,17);
30 N_min = SOLAR_MIN(:,18);
31
32 % NRLMSIS uses 0 or 1e-38 for small densities, filter these out
33 O_max(O_max < 1e-30) = NaN;
34 N2_max(N2_max < 1e-30) = NaN;
35 O2_max(O2_max < 1e-30) = NaN;
36 total_mass_max(total_mass_max < 1e-30) = NaN;
37 He_max(He_max < 1e-30) = NaN;
38 Ar_max(Ar_max < 1e-30) = NaN;
39 H_max(H_max < 1e-30) = NaN;
40 N_max(N_max < 1e-30) = NaN;
41
42 O_min(O_min < 1e-30) = NaN;
43 N2_min(N2_min < 1e-30) = NaN;
44 O2_min(O2_min < 1e-30) = NaN;
45 total_mass_min(total_mass_min < 1e-30) = NaN;
46 He_min(He_min < 1e-30) = NaN;
47 Ar_min(Ar_min < 1e-30) = NaN;
48 H_min(H_min < 1e-30) = NaN;
49 N_min(N_min < 1e-30) = NaN;
50
51 colors = hsv(8);
52
53 figure
54 set(gca, 'YScale', 'log'); % plot as log plot
55 hold on

```

```

56 semilogy(alt_max_km, O_max, '-', 'Color', colors(1,:), 'LineWidth', 1.5, 'DisplayName', 'O - Solar
    ↪ Max')
57 semilogy(alt_min_km, O_min, '--', 'Color', colors(1,:), 'LineWidth', 1.5, 'DisplayName', 'O - Solar
    ↪ Min')
58 semilogy(alt_max_km, N2_max, '-', 'Color', colors(2,:), 'LineWidth', 1.5, 'DisplayName', 'N_2 -
    ↪ Solar Max')
59 semilogy(alt_min_km, N2_min, '--', 'Color', colors(2,:), 'LineWidth', 1.5, 'DisplayName', 'N_2 -
    ↪ Solar Min')
60 semilogy(alt_max_km, O2_max, '-', 'Color', colors(3,:), 'LineWidth', 1.5, 'DisplayName', 'O_2 -
    ↪ Solar Max')
61 semilogy(alt_min_km, O2_min, '--', 'Color', colors(3,:), 'LineWidth', 1.5, 'DisplayName', 'O_2 -
    ↪ Solar Min')
62 semilogy(alt_max_km, total_mass_max, '-', 'Color', colors(4,:), 'LineWidth', 1.5, 'DisplayName', '
    ↪ Total Mass Density - Solar Max')
63 semilogy(alt_min_km, total_mass_min, '--', 'Color', colors(4,:), 'LineWidth', 1.5, 'DisplayName', '
    ↪ Total Mass Density - Solar Min')
64 semilogy(alt_max_km, He_max, '-', 'Color', colors(5,:), 'LineWidth', 1.5, 'DisplayName', 'He -
    ↪ Solar Max')
65 semilogy(alt_min_km, He_min, '--', 'Color', colors(5,:), 'LineWidth', 1.5, 'DisplayName', 'He -
    ↪ Solar Min')
66 semilogy(alt_max_km, Ar_max, '-', 'Color', colors(6,:), 'LineWidth', 1.5, 'DisplayName', 'Ar -
    ↪ Solar Max')
67 semilogy(alt_min_km, Ar_min, '--', 'Color', colors(6,:), 'LineWidth', 1.5, 'DisplayName', 'Ar -
    ↪ Solar Min')
68 semilogy(alt_max_km, H_max, '-', 'Color', colors(7,:), 'LineWidth', 1.5, 'DisplayName', 'H - Solar
    ↪ Max')
69 semilogy(alt_min_km, H_min, '--', 'Color', colors(7,:), 'LineWidth', 1.5, 'DisplayName', 'H - Solar
    ↪ Min')
70 semilogy(alt_max_km, N_max, '-', 'Color', colors(8,:), 'LineWidth', 1.5, 'DisplayName', 'N - Solar
    ↪ Max')
71 semilogy(alt_min_km, N_min, '--', 'Color', colors(8,:), 'LineWidth', 1.5, 'DisplayName', 'N - Solar
    ↪ Min')
72 grid on
73 xlabel('Altitude (km)')
74 ylabel('Density (g/cm^3)')
75 title('NRLMSIS Species Density Profiles: Solar Max vs Solar Min')
76 legend('Location', 'eastoutside')
77
78 set(gcf, 'Units', 'inches', 'Position', [0 0 12 4]) % smaller physical size for the doc
79 exportgraphics(gcf, 'NRLMSIS_Species_Density_Profiles.png', 'Resolution', 600);

```

```

1  %% ASEN 5335 Aerospace Environment
2  %% Justin Le
3  %% HW2 Problem 5
4
5  clc;clear; close all
6
7  % Constants
8  G = 6.67e-11;
9  M_E = 5.972e24;
10 R_E_m = 6371e3;
11 C_D = 2.2;
12 m_sat_kg = 4; % From earlier in notes
13 A_x_m2 = 0.09;
14 A_z_m2 = 0.01;
15 stop_alt_km = 120;
16
17 % Part ii, 400 km altitude in +x direction
18 [t2_days, alt2_km] = simulateDecay(400, A_x_m2, stop_alt_km, G, M_E, R_E_m, C_D, m_sat_kg);
19 % Part iii, 500 km in +x direction
20 [t3_days, alt3_km] = simulateDecay(500, A_x_m2, stop_alt_km, G, M_E, R_E_m, C_D, m_sat_kg);
21 % Part iv, 400 km altitude in +z direction
22 [t4_days, alt4_km] = simulateDecay(400, A_z_m2, stop_alt_km, G, M_E, R_E_m, C_D, m_sat_kg);
23
24 figure
25 plot(t2_days, alt2_km)
26 grid on
27 xlabel('Time (days)')
28 ylabel('Altitude (km)')
29 title('Orbital Decay: 400 km Start, Ram in +X')
30 set(gcf, 'Units', 'inches', 'Position', [0 0 6 4])
31 exportgraphics(gcf, 'HW2Problem5_OrbitDecay_400km_x.png', 'Resolution', 600);
32
33 figure
34 plot(t3_days, alt3_km)
35 grid on
36 xlabel('Time (days)')
37 ylabel('Altitude (km)')
38 title('Orbital Decay: 500 km Start, Ram in +X')
39 set(gcf, 'Units', 'inches', 'Position', [0 0 6 4])
40 exportgraphics(gcf, 'HW2Problem5_OrbitDecay_500km_x.png', 'Resolution', 600);
41
42 figure
43 plot(t4_days, alt4_km)
44 grid on
45 xlabel('Time (days)')
46 ylabel('Altitude (km)')
47 title('Orbital Decay: 400 km Start, Ram in +Z')
48 set(gcf, 'Units', 'inches', 'Position', [0 0 6 4])
49 exportgraphics(gcf, 'HW2Problem5_OrbitDecay_400km_z.png', 'Resolution', 600);
50
51 function [t_days, alt_km] = simulateDecay(alt0_km, A_m2, stop_alt_km, G, M_E, R_E_m, C_D, m_sat)
52 % Calculate orbital elements at this time point
53 a = R_E_m + alt0_km*1000;
54 P = 2*pi*sqrt(a^3/(G*M_E));
55
56 % Conditions to iterate
57 t = 0;
58 altitude_index = 1;
59 t_hist(altitude_index) = 0;

```

```
60 alt_hist(altitude_index) = alt0_km;
61
62 % At start, alt = alt0.
63 alt_current_km = alt0_km;
64 % Loop until sat deorbits
65 while alt_current_km > stop_alt_km
66     % Find density at that altitude
67     MSIS = MSISatmosphere1000(alt_current_km);
68     rho = MSIS.mass*1000;
69
70     % Equation 10 and find change in Period
71     dPdt = -3*pi*a*rho*C_D*A_m2/m_sat;
72
73     dt = 600;
74
75     P = P + dPdt*dt;
76     t = t + dt;
77
78     a = (G*M_E*P^2/(4*pi^2))^(1/3);
79     alt_current_km = (a - R_E_m)/1000
80
81     altitude_index = altitude_index + 1;
82     t_hist(altitude_index) = t/86400; % Convert to days
83     alt_hist(altitude_index) = alt_current_km;
84 end
85
86 t_days = t_hist;
87 alt_km = alt_hist;
88
89 end
```